

UBC CPSC 340 2019W1

MIDTERM EXAM

Oct 17th, 2018

Instructor: Mark Schmidt

TIME: 80 minutes

We are providing a copy of this exam to help you prepare for the style of questions we may ask during the midterm and final. However, the solution file is meant for you alone and we do not give permission to share these solution files with anyone. Both distributing solution files to other people or using solution files provided to you by other people are considered academic misconduct. Please see UBC's policy on this topic if you are not familiar with it:

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- Do not open the exam until you are directed to do so.
 - Once you open the exam, make sure that it contains this cover page plus 8 pages of exam questions.
 - One letter-size sheet (both sides) of notes is allowed. No other material or accessories may be used.
 - You may use either pen or pencil, but exams written in pencil may not be eligible for regrading.
 - Please be prepared to present, upon request, a student card for identification.
 - If you need more space, use the blank page at the end of the exam, and clearly indicate that your work continues there.
 - Most questions require a short answer. Work efficiently and avoid writing lengthy answers.
 - The exam uses the notation from class (n refers to the number of training examples, d is the number of features, and so on).
 - If anything is unclear or seems ambiguous, **state your assumptions**.
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Question:	1	2	3	4	5	6	7	Total
Points:	10	15	6	6	8	6	9	60
Score:								

Question 1.

(10 points)

6 pts

- (a) Using the notation from class, **give the size** of the following quantities in terms of n , d , and t . And in one short sentence, **describe what the notation represents** in this course. For example, for y you could write “ $n \times 1$: label of training example i ”.

i. x_{ij} ii. x_i iii. $\hat{y}_i$ iv. $\tilde{y}_i$ v. X vi. $\tilde{X}$

4 pts

- (b) Which of the following methods are examples of **supervised learning**? Circle all that apply.

- i. Hierarchical clustering
- ii. KNN classification
- iii. k -medians
- iv. Linear regression
- v. Naive Bayes
- vi. Outlierness ratio
- vii. Random forests
- viii. Robust regression

Question 2.

(15 points)

5 pts

- (a) Which of the following methods are **non-parametric**? Circle all that apply.
- i. Random forests with depth-20 random trees
 - ii. k -means clustering using $k = \log(n)$.
 - iii. Density-based clustering with $\epsilon = 0.1$, $\text{minNeighbours} = 10$.
 - iv. Density-based clustering with $\epsilon = 0.1$, $\text{minNeighbours} = \log(n)$.
 - v. Linear regression with degree $p = 7$ polynomial basis.

10 pts

- (b) Which of the following changes would typically **reduce training error**? Circle all that apply.

Note: for regression and unsupervised models, *assume we use the squared training error* (squared distance to cluster mean for k -means, and squared prediction error for regression).

- i. using the fast decision stump fitting method based on sorting, instead of the naive approach
- ii. increasing k in KNN classification
- iii. increasing the maximum tree depth in a random forest
- iv. increasing k in k -means clustering
- v. using a higher degree polynomial basis for linear regression
- vi. running more iterations of gradient descent when fitting least squares (assuming α is small enough)
- vii. adding some relevant features to your model
- viii. adding some irrelevant features to your model
- ix. switching from the squared error to the absolute error when fitting a linear regression model
- x. increasing the parameter ϵ in the Huber loss

Question 3.**(6 points)**

Answer the questions below using 1-2 short sentences.

2 pts

- (a) What is a setting where using a validation set to choose hyper-parameters can lead to overfitting?

2 pts

- (b) What is the difference between using a validation set and using cross-validation?

2 pts

- (c) Consider an ensemble clustering method that generates m different bootstraps of the data. It then fits a k-means model (with a random initialization) to each of the bootstraps. To form the final clustering for example x_i , it chooses the y_i that is most common across the m clusterings. Would this be an effective or an ineffective ensemble method? (Briefly explain.)

Question 4.

(6 points)

Answer the questions below using 1-2 short sentences.

2 pts

- (a) Suppose we have a supervised learning problem where we think the examples x_i form clusters. To deal with this, we combine our training and test data together and cluster the combined data using k-means. We then add the cluster number as an extra feature, fit our supervised learning model based on the training data, then evaluate it on the test data. What have we done wrong?

2 pts

- (b) We considered fitting decision stumps by finding the stump that maximizes the number of times $\hat{y}_i = y_i$ in the training data. Why do we not try to maximize this quantity when we fit the regression weights w in linear regression models?

2 pts

- (c) We argued that penalizing the L1-norm of the residuals makes linear regression less sensitive to outliers. This is also true of the Huber loss, so why did we introduce the Huber loss?

Question 5.

(8 points)

Answer the following questions using **big-O** notation. Your answers may involve n , d , and perhaps additional quantities defined in the question. As an example, (linear) least squares has $O(d)$ parameters, requires $O(nd^2 + d^3)$ time to train, and requires $O(td)$ time to predict on t test examples.

2 pts

- (a) What is the storage space required for a naive Bayes classifier with binary features and k class labels?

2 pts

- (b) What is the prediction cost for classifying t test examples using a random forest classifier, if each tree has a maximum depth of k and we use m trees?

2 pts

- (c) What is the training time for linear regression with $d = 1$ using a polynomial basis of degree p ?

2 pts

- (d) What is the cost of computing the gradient of the Huber loss function? Recall that the gradient of the Huber loss has the form $\nabla f(w) = \sum_{i=1}^n h'(w^T x_i - y_i) x_i$, where it costs $O(1)$ to evaluate $h'(r_i)$ for a given scalar r_i .

Question 6.**(6 points)**

Consider the dataset below, which has 10 training examples and 2 features:

$$X = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 1 \\ 1 & 0 \\ 1 & 0 \\ 1 & 0 \\ 1 & 1 \\ 1 & 1 \\ 1 & 1 \\ 1 & 1 \end{bmatrix}, \quad y = \begin{bmatrix} -1 \\ -1 \\ -1 \\ +1 \\ +1 \\ -1 \\ +1 \\ +1 \\ +1 \\ -1 \end{bmatrix}.$$

Suppose you want to classify the following test example:

$$\tilde{x}_i = \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

2 pts

(a) What value of y_i would a 4-nearest neighbour classifier predict for this test example?

2 pts

(b) If we treated the y_i as numerical values, what value of y_i would a linear regression model predict on the test example $\tilde{x}_i$ if it had the below weights?

$$w = \begin{bmatrix} 1.5 \\ -2.3 \end{bmatrix}$$

2 pts

(c) If we fit a decision stump to maximize the classification accuracy score, what stump would we choose?

Question 7.

(9 points)

For this question you may find it helpful to use the next page, which is blank.

3 pts

- (a) Consider the following objective, which considers a weighted absolute error with a penalty on the maximum coordinate-wise deviation away from some target value w^0 (that has elements $w_1^0, w_2^0, \dots, w_d^0$),

$$f(w) = \sum_{i=1}^n v_i |w^T x_i - y_i| + \lambda \max_{j \in \{1, 2, \dots, d\}} |w_j - w_j^0|,$$

where λ is a non-negative scalar. Re-write this objective function in matrix and norm notation. You can use V as a diagonal matrix with the (non-negative) elements v_i along the diagonal. Make sure your dimensions match.

3 pts

- (b) Consider an L2-regularized least squares objective, with a non-diagonal regularizer,

$$f(w) = \frac{1}{2} \|Xw - y\|^2 + \frac{1}{2} w^T \Lambda w,$$

where Λ is a symmetric matrix. Write down a linear system whose solution gives a stationary point of this objective function. Make sure your dimensions match.

3 pts

- (c) Which of the following functions are **convex**? Circle all that apply.

- i. $f(w) = \sum_{i=1}^n \lambda_i w^T x_i$ (where each scalar $\lambda_i > 0$).
- ii. $f(w) = \max\{w, w^3\}$.
- iii. $f(w) = \|Xw - y\|_3 + \lambda \|w\|_1$
(where $\lambda > 0$ and the “3-norm” is $\|r\|_3 = (\sum_{i=1}^n |r_i|^3)^{1/3}$, which is a norm).

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